

Tiger Lu

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APPOINTMENTS	Center for Computational Astrophysics , Flatiron Institute, New York, NY Flatiron Research Fellow	Sep 2025 – Present
	Yale University , New Haven, CT Graduate Research Assistant	Aug 2020 – May 2025
	University of California, Berkeley , Berkeley, CA Summer Undergraduate Researcher	May 2019 – Aug 2019
	Carnegie Observatories , Pasadena, CA Summer Undergraduate Researcher	May 2018 – Aug 2018

EDUCATION	Yale University , New Haven, Connecticut, USA	
	▪ Ph.D. in Astronomy with Distinction	Aug 2020 – May 2025
	• Thesis: Planets are not Points: the Profound Effect of Planetary Structure on Exoplanet System Architectures	
	• Advisers: Prof. Gregory Laughlin, Prof. Malena Rice	
	▪ M.S., M.Phil. in Astronomy	Aug 2020 – Aug 2022
	California Institute of Technology , Pasadena, California, USA	
	▪ B.S. in Astrophysics	Aug 2016 – Jun 2020
	▪ Minor in Computer Science	

PUBLICATIONS 188 citations. h -index = 8.
Italics: Student directly supervised by T. Lu

FIRST-AUTHOR

- [9] **Lu, T.**, Millholland, S.C., Rice, M., Black, B., Bardalez Gagliuffi, D., Balmer, W., Pueyo, L., Giovannazzi, M., and Brandt, T. In review.
- [8] **Lu, T.**, Hernandez, D.M., Rice, M. and Su, Y. In review.
- [7] **Lu, T.** and Brown, Garrett. [A Coordinate System for Dynamical Instabilities in Hierarchical Systems in REBOUND](#). In review.
- [6] **Lu, T.**, Tajar, H., Hernandez, D.M., Rein, H., *Liu, Y.* and Rice, M., 2025. [Collisional Fragmentation Support in TRACE](#). *RNAAS* **9**, 5
- [5] **Lu, T.**, Li, G., Cassese, B. and Lin, D.N.C., 2025. [The Dynamical History of HIP-41378 f - Oblique Exorings Masquerading as a Puffy Planet](#). *ApJ* **980**, 39
- [4] **Lu, T.**, An, Q., Li, G., Millholland, S.C., Rice, M., Brandt, G.M. and Brandt, T., 2025. [Planet-Planet Scattering and ZLK Migration – the Dynamical History of HAT-P-11](#). *ApJ* **979**, 218
- [3] **Lu, T.**, Hernandez, D.M. and Rein, H., 2024. [TRACE: a Time-Reversible Algorithm for Close Encounters](#). *MNRAS* **533**, 3708
- [2] **Lu, T.**, Rein, H., Tamayo, D., Hadden, S., Mardling, R., Millholland, S.C. and Laughlin, G., 2023. [Self-Consistent Spin, Tidal and Dynamical Equations of Motion in the REBOUNDx Framework](#). *ApJ* **948**, 41
- [1] **Lu, T.** and Laughlin, G., 2022. [Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9](#). *PSJ* **3**, 221

STUDENT-LEAD

- [4] *Yang, E.*, **Lu, T.**, Yahalom, D., and Winn, J. In review.
- [3] *Im, H.*, **Lu, T.**, Rice, M., Tran, Q., Li, G. and Naoz, S., 2026. [Observational and Dynamical Constraints on an Unseen Outer Perturber in the GJ 436 Hot Neptune System](#). *ApJ* **1003**, 84
- [2] *Liu, Y.*, **Lu, T.** and Rice, M., 2025. [Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94](#). *ApJL* **997**, L41

- [1] [Liu, Y., Lu, T. and Rice, M., 2025. The Formation of Double Hot Jupiter Systems Through ZLK Migration. *ApJ* **986**, 103](#)

SIGNIFICANT CONTRIBUTION

- [6] Rescigno, F., Stalport, M., Anna John, A., **Lu, T.**, et al. in review
- [5] Yahalomi, D.A., **Lu, T.**, Armitage, P., Bedell, M., Casey, A.R., Price-Whelan, A., and Rice, M., 2026. [The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry. *ApJL* **999**, L9](#)
- [4] Cassese, B., Rice, M. and **Lu, T.**, 2025. [A High-Precision, Differentiable Code for Solar System Ephemerides. *PSJ* **6**, 252](#)
- [3] An, Q., **Lu, T.**, Brandt, G.M., Brandt, T. and Li, G., 2025. [Significant Mutual Inclinations Between the Stellar Spin and the Orbits of Both Planets in the HAT-P-11 System. *AJ* **169**, 22](#)
- [2] Cassese, B., Vega, J., **Lu, T.**, Rice, M., Poddar, A. and Kipping, D., 2024. [squishyplanet: Modeling transits of non-spherical planets in JAX. *JOSS* **9**, 100](#)
- [1] Wahl, S., Thorngren, D., **Lu, T.** and Militzer, B., 2021. [Tidal Response and Shape of Hot Jupiters *ApJ* **921**, 105](#)

OTHER CO-AUTHOR

- [3] Tajer, H., Wang, J., Childs, A., et al. (inc. **Lu, T.**) 2026. [Origins of Mercury’s Big Heart of Iron: Exploring Pathways to Form High Core Mass Fraction \(CMF\) Planets via N-body Simulations. *ApJ* **998**, 46](#)
- [2] Bardalez Gagliuffi, D.†, Balmer, W.†, Pueyo, L., et al. (inc. **Lu, T.**) 2025. [JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet in a Dynamically Hot, Multi-planet System. *ApJL* **988**, L18](#)
- [1] Wang, N., Lu, L.Y., Liu, H.G., et al. (inc. **Lu, T.**) 2025. [Development of Gravity Theories in the View of TRAPPIST-1e. *RAA* **25**, 015003](#)

WHITEPAPERS

- [1] Levine, W. G., Gerbig, K., Loudon, E., **Lu, T.**, Hsieh, C., O’Connor, C., Li, R. and Dong, J., 2024. [Emerging Researchers in Exoplanetary Science \(ERES\): Lessons Learned in Conference Organization for Early-Career Researchers. Bulletin of the AAS **56**, 2024i004](#)

†: Joint first author

AWARDS & FELLOWSHIPS

▪ DDA/AAS Raynor L. Duncombe Student Research Prize	2025
▪ Flatiron Research Fellowship	2025
▪ Flipped Science Fair Best Poster Explanation	2024
▪ AAS International Travel Grant	2024
▪ Graduate Student Assembly Conference Travel Fellowship	2024
▪ National Science Foundation GRFP (<i>Honorable Mention</i>)	2022
▪ Joan and Arnold Seidel Griffith Observer Science Writing Contest (<i>Second Place</i>)	2022
▪ Alan Porter Memorial Foundation Fellow	2019
▪ Summer Undergraduate Research Fellow	2018

ORAL PRESENTATIONS & TALKS

▪ New York Area Exoplanets Meeting III, AMNH (<i>New York City, NY</i>)	May 2026
▪ Monday Afternoon Talk, MIT (<i>Cambridge, MA</i>)	Apr 2026
▪ Astronomy & Planetary Science Colloquium*, NAU (<i>Flagstaff, AZ</i>)	Feb 2026
▪ Astronomy Seminar*, AMNH (<i>New York City, NY</i>)	Feb 2026
▪ International Conference on Exoplanets and Planet Formation (<i>Shanghai, China</i>)	Dec 2025
▪ Powell Group Meeting, University of Chicago (<i>Chicago, IL</i>)	Oct 2025
▪ 246th AAS Meeting Dissertation Talk (<i>Anchorage, AK</i>)	Jun 2025
▪ Boston Area Planetary Science Meeting, MIT (<i>Cambridge, MA</i>)	May 2025
▪ New York Area Exoplanets Meeting II, CCA (<i>New York City, NY</i>)	May 2025
▪ 56th DDA Meeting, Georgia Institute of Technology (<i>Atlanta, GA</i>)	May 2025
▪ ML in Planetary Dynamics Workshop*, CCA (<i>New York City, NY</i>)	Apr 2025
▪ Tea Talk, Indiana University (<i>Bloomington, IN</i>)	Feb 2025
▪ ITC Luncheon, Harvard University (<i>Cambridge, MA</i>)	Jan 2025
▪ Exoplanet Pizza Lunch, Harvard University (<i>Cambridge, MA</i>)	Jan 2025

	▪ Exoplanet Seminar, Princeton University (<i>Princeton, NJ</i>) ▪ Planetary Lunch Seminar, Massachusetts Institute of Technology (<i>Cambridge, MA</i>) ▪ Astronomy Seminar, Columbia University (<i>New York City, NY</i>) ▪ Astronomy Seminar, Stony Brook University (<i>Stony Brook, NY</i>) ▪ CRA Seminar, Georgia Institute of Technology (<i>Atlanta, GA</i>) ▪ Planetary Science Seminar, California Institute of Technology (<i>Pasadena, CA</i>) ▪ Lunch Seminar, UCLA (<i>Los Angeles, CA</i>) ▪ Exoplanet Group Meeting, UCSB (<i>Santa Barbara, CA</i>) ▪ FLASH Talk, UCSC (<i>Santa Cruz, CA</i>) ▪ Star and Planet Lunch, University of Hawai'i (<i>Honolulu, HI</i>) ▪ CT Exoplanet Picnic, Wesleyan College (<i>Wesleyan, CT</i>) ▪ Emerging Researchers in Exoplanet Science IX, Cornell University (<i>Ithaca, NY</i>) ▪ REBOUND 24 Meeting* (<i>Virtual</i>) ▪ <i>Arbitrary ODE Workshop</i> ▪ REBOUND 24 Meeting* (<i>Virtual</i>) ▪ <i>tides_spin</i> ▪ New York Area Exoplanets Meeting I, CCA (<i>New York City, NY</i>) ▪ 55th DDA Meeting, University of Toronto (<i>Toronto, Canada</i>) ▪ Makino Lab Seminar, Kobe University (<i>Kobe, Japan</i>) ▪ Exoplanet Seminar, Nanjing University (<i>Nanjing, China</i>) ▪ 54th DDA Meeting, Michigan State University (<i>East Lansing, MI</i>) ▪ Exoplanet Seminar, Princeton University (<i>Princeton, NJ</i>) ▪ Astronomy Seminar, Monash University (<i>Monash, Australia</i>) ▪ 240th AAS Meeting (<i>Pasadena, CA</i>)	Nov 2024 Nov 2024 Oct 2024 Oct 2024 Oct 2024 Oct 2024 Sep 2024 Sep 2024 Sep 2024 Aug 2024 Jul 2024 Jul 2024 Jul 2024 Jul 2024 May 2024 May 2024 Jul 2023 Jul 2023 May 2023 Mar 2023 Feb 2023 Jun 2022
	*Invited Talk	
POSTER PRESENTATIONS	▪ Emerging Researchers in Exoplanet Science, Princeton University (<i>Princeton, NJ</i>) ▪ Exoplanets V Meeting (<i>Leiden, The Netherlands</i>) ▪ Extreme Solar Systems (<i>Christchurch, New Zealand</i>) ▪ 2022 AGU Meeting (<i>Chicago, IL</i>) ▪ Exoplanets IV Meeting (<i>Las Vegas, NV</i>) ▪ 2019 AGU Meeting (<i>San Francisco, CA</i>) ▪ 233rd AAS Meeting (<i>Seattle, Washington</i>)	Jun 2025 Jun 2024 Jun 2024 Dec 2022 May 2022 Dec 2019 Jan 2019
TEACHING	ASTR 210: Stars and Their Evolution Yale University Teaching Fellow ASTR 160: Frontiers and Controversies in Astrophysics Yale University Teaching Fellow ASTR 160: Frontiers and Controversies in Astrophysics Yale University Teaching Fellow ASTR 155: Introduction to Astronomical Observing Yale University Teaching Fellow Dean's Tutor Tutoring Caltech students in Physics, Astronomy and Computer Science	New Haven, CT, USA Spring 2023 New Haven, CT, USA Fall 2022 New Haven, CT, USA Spring 2021 New Haven, CT, USA Fall 2021 Pasadena, CA, USA 2017-2020
OUTREACH	Warrior-Scholar Program Research Project Leader <i>One week intensive boot camp preparing military veterans for undergraduate degrees</i> Yale OpenLabs Local outreach organization dedicated to educating middle school children on STEM topics. Receptient of the Seton Elm-Ivy Award. ▪ Public Relations Officer ▪ Regular Outreach Speaker • Science Cafe 2024 Speaker: <i>What if 'Cotton Candy' Planet are Actually Ring Pops?</i> • Flipped Science Fair Best Poster Explanation: <i>Probing the Shapes of Planets</i>	New Haven, CT, USA Summer 2023 2023 2022 – 2025

- Science Cafe 2022 Speaker: [How We Discover Exoplanets](#)

Astronomy on Tap

New Haven branch of outreach program aimed at engaging local community with astronomy research

- Social Media Director 2022

Science in the News

Local outreach organization dedicated to educating the greater community on STEM topics.

- Outreach Speaker 2021
 - [Hidden Things: Pulling Back the Veil on the Universe](#)

RESEARCH MENTORING

Yurou (Nina) Liu*

Alexander P. Hixon Summer Undergraduate Research Fellow

Project: *The Formation of Double Hot Jupiter Systems Through ZLK Migration*

Jun 2023 –
Yale University

Haedam Im

Dorrit Hoffleit Undergraduate Research Fellow

Project: *Hunting for a Hidden Planet in the GJ 436 System*

May 2025 – Apr 2026
Massachusetts Institute of Technology

Eritas Yang

Graduate Student

Oct 2025 –
Princeton University

Vighnesh Nagpal

Graduate Student

Aug 2025 –
University of Chicago

Haniyeh Tajer

Graduate Student

May 2025 –
Ohio State University

Jeremiah Reynoso

Summer Undergraduate Research Fellow

*Primary Research Advisor

Jun 2023 – Oct 2023
Morehouse College

INFORMAL MENTORING

Chris Santiago

Astro Sibb

2022 – 2024
Yale University

Kylyn Smith

Astro Sibb

2023 – 2024
Yale University

Sally Jiang (current Ph.D. Candidate, Columbia University)

Astro Sibb

2022 – 2023
Yale University

TELESCOPE PROPOSALS (CO-I)

James Webb Space Telescope – 44.7 hours

ID: 11253. Fluffy or Flashy? Resolving the Mystery of the Coldest-Known Super Puff. *PI: Ben Cassese*

Cycle 5 GO

Keck Observatory (KPF) – 0.5 nights

Directly Probing the Migration History of WASP-53 with Precise Orbital Constraints. *PI: Quang Tran*

Yale 2026A

Keck Observatory (KPF) – 2 nights

The Search for Double Hot Jupiter Systems. *PI: Malena Rice*

Yale 2024B

Keck Observatory (HIRES) – 2 nights

Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 2). *PI: Emma Louden*

Yale 2023A

Keck Observatory (HIRES) – 2 nights

Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 1). *PI: Emma Louden*

Yale 2022B

PROFESSIONAL SERVICE

Science Organizing Committee – 57th DDA Meeting

Division of Dynamical Astronomy, University of Chicago

June 2026

Science Organizing Committee Chair – DAPS

A Dynamical Synthesis of Planetary Systems, New Haven

Jan 2026

Organizing Committee – ERES X

Emerging Researchers in Exoplanet Science Symposium, Princeton University

July 2025

Science Organizing Committee – ERES VIII

June 2023

Emerging Researchers in Exoplanet Science Symposium, Yale University

Reviewer

NASA XRP, NASA Solar System Workings Panel, A&A, AJ, ApJ, MNRAS

2024 – Present

Main Co-Organizer – Yale Exoplanets and Stars Seminar

2023 – 2025

Weekly Seminar Series

Committee Member – ACDC

2022 – 2025

Yale University Astronomy Climate and Diversity Committee

MEDIA

Yale Daily News

Jun 2025

[New study offers a double dose of ‘hot Jupiters’](#)

Popular Science

Oct 2024

[What if ‘cotton candy’ planets are actually Ring Pops?](#)

AAS Author Journal Series

Nov 2022

[Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9](#)

**COMMUNITY
ENGAGEMENT**

New Haven Road Runners – New Haven, CT, USA

Local USATF sanctioned running club

- Public Relations Officer
- Racing Team Director / Elite Athlete Coordinator

Jul 2023 – Jul 2024

Jul 2022 – Jun 2025

NCAA Athlete – Pasadena, CA, USA

2016 – 2020

NCAA Division III Cross Country/Track & Field at Caltech

Avery House – Pasadena, CA, USA

Student Government

- Athletics Manager
- Underclassmen Counselor

2017 – 2018

2017 – 2020

[CV compiled on 2026-05-26]